Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/01/2014
PI Name:	O'Brian, Thomas Ph.D.		
Project Title:	NIST/ACES Collaboration: ACES Ground Station and Data Analysis Center in Boulder, Colorado		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	chris.oates@nist.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	303-497-4570
Organization Name:	National Institute of Standards and Technology		
PI Address 1:	Time and Frequency Division		
PI Address 2:	325 Broadway Street, MS 847.00		
PI Web Page:			
City:	Boulder	**State:**	CO
Zip Code:	80305-3337	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	06/01/2012	**End Date:**	11/15/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
Contact Email:	ulf.e.israelsson@jpl.nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed in November 2015 due to change in PI, per PI and new PI (Ed., 12/17/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Jefferts, Steven (National Institute of Standards & Technology)		
Grant/Contract No.:	NNH12AT81I		
Performance Goal No.:			
Performance Goal Text:			

To support primary ACES mission objectives, NIST proposes to:
1. Develop a Microwave Link Ground Terminal (MWL GT) at NIST in Boulder, CO to perform 10-17 uncertainty time and frequency comparisons between ACES clocks and world-leading optical and atomic clocks at NIST Boulder and the nearby JILA; and to support general ACES science goals.

2. Participate as fully as possible in coordinated ESA data analysis with formal ESA Data Analysis Centers and other ACES partners throughout the world. NIST will collaborate closely with ESA partners to define how NIST can best contribute to overall ACES mission goals within the constraints of NIST resources.

As part of participation in the time and frequency transfer and data analysis, NIST will participate in tests of fundamental

Task Description:	physics based on ACES data to include at least the following: • Global time and frequency transfer with uncertainty as low as 10-17. • Direct comparison of PHARAO space clock to NIST time and frequency standards at uncertainty of 10-17 or better. • Test of gravitation redshift at 2 x 10-6 uncertainty or better. • Search for variation in fundamental constants, such as fine structure constant at level of 10-18 per year or better. All these goals will be facilitated by the strong time and frequency infrastructure at NIST, including (a partial list): • World-class cesium microwave primary frequency standards, including NIST-F1 with fractional frequency uncertainty of 3 x 10-16, and NIST-F2 which will be operational in 2013 with an eventual uncertainty of 1 x 10-16 or better. • Two mercury ion optical clocks with current fractional frequency uncertainty of approximately 1 x 10-17 and continually improving. • Two aluminum ion “logic” optical clocks with current fractional frequency uncertainty of approximately 8 x 10-18 and continually improving. • Two ytterbium lattice optical clocks with uncertainties better than 1 x 10-16 and rapidly improving. • Two strontium lattice optical clocks with uncertainties approximately 3 x 10-17 and rapidly improving. (At JILA. A dedicated fiber optic system connects NIST and JILA to enable the JILA clocks to directly participate in NIST time and frequency transfer and intercomparisons.) • A high performance calcium optical frequency standard with exquisite short-term stability to serve as a “fly wheel” for clock comparisons. • A world-class time scale comprising seven hydrogen masers, with eight more masers to be added over the next three years. NIST scientists will participate in ESA/NASA planning meetings about the ACES program in general, and will specifically consult with ESA to determine how NIST can most fully participate in MLW GT data analysis and applications. In addition to the NIST resources already proposed for this program (NIST and JILA optical and microwave frequency standards, NIST time scale, NIST time and frequency transfer capabilities, etc.) NIST will directly contribute substantial funds and expert NIST staff time significantly exceeding the direct NASA investment. We propose the following revised program plan outline: Year 1 (2012): Begin design and construction of platform and other infrastructure for MWL hardware. $150k for partial payment of platform and related infrastructure design and construction. Year 2 (2013): Complete construction of platform and other infrastructure for MWL system. $150k for partial payment of platform and related infrastructure. Year 3 (2014): Receive MWL hardware. Install, test, and integrate MWL hardware into NIST time and frequency transfer system. Prepare and test MWL system for ACES time and frequency transfer. New NIST staff to prepare for participation in MWL data analysis. $150k for partial support of staff time. Year 4 (2015): ACES launch. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time. Year 5 (2016): On-going ACES experiments. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The major benefits and impacts of this research program can be grouped into three main areas: 1. Substantial improvements in international coordination of extremely precise and accurate time and frequency information. Precision timekeeping and synchronization are crucial for such ubiquitous technologies as precision positioning and navigation, telecommunications, remote sensing, surveillance, and many other applications. The current technical limitations on international time and frequency transfer impose fundamental limits on these technologies. This project seeks to improve international time and frequency transfer by at least a factor of 10, with the potential for significantly greater improvements. Such improvements would not only directly enhance the current technologies depending on precision timing and synchronization, but would also open up new applications directly related to ultraprecise timing such as gravimetry, geodesy, magnetometry, and other measurements based on tiny perturbations in precision frequency measurements. 2. This project will serve as a pathfinder demonstration of the ability to perform cold atom experiments in space. Cold atoms and molecules represent a major, growing research opportunity for a wide variety of technologies, from ultraprecision measurements of all kinds (time, gravity, magnetic fields, electrical quantities, temperature, etc.) to the potential for development of exotic new materials. Having mastered cold atom and molecule experiments in the earth laboratory, being able to perform such experiments in space opens the benefits of precision measurements in the space environment as well as microgravity experiments not possible in the Earth-based laboratory. 3. Fundamental physics experiments. The operation of a cold atom clock in space provides the opportunity to significantly improve measurements on the limits of special and general relativity and related phenomena, such search for possible Lorentz transformation variations and search for possible time variations in fundamental constants, at levels 10 to 100 times more precise than previous experiments. Most major physics discoveries in recent decades have come from tiny variations in expected behavior, and these experiments would have profound implications on fundamental physics and cosmology.

Task Progress:

In the most recent reporting period (April 2013 through March 2014) work has focused on three main tasks:
1. Preparation of the NIST site for the Microwave Link Ground Terminal (MWL GT).

2. Advancing the NIST time and frequency metrology infrastructure in support of ACES goals for precision time and frequency transfer and tests of fundamental physics.

3. Theory support for ACES goals for precision time and frequency transfer and tests of fundamental physics.

Progress on each task is described below.

1. Preparation of the NIST site for the Microwave Link Ground Terminal (MWL GT). The Microwave Link Ground Terminal (MWL GT) system will be provided by ESA to participating ground station partners across the world, including NIST in Boulder, Colorado. The MWL GT systems are still under development through ESA sponsorship. NIST has been closely coordinating with ESA contacts on details of the MWL GT hardware, information technology and infrastructure requirements.

NIST scientists worked with NIST facilities staff and engineering firms to develop the best location for the MWL GT that provides the best available views of the ISS as it passes by in orbit, and to develop the physical and support infrastructure needed to site the MWL GT on the roof of a NIST building, provide required utilities and information technology support, and provide other required infrastructure. The optimal site was located on the roof of the new $100 million NIST Precision Measurement Laboratory, providing the best view of the ISS, the best infrastructure to support the MWL GT hardware, and providing a location for easy access to signals from NIST microwave and optical frequency standards. A design and construction contract was awarded for the platform and infrastructure to support the MWL GT.

2. Advancing the NIST time and frequency metrology infrastructure in support of ACES goals for precision time and frequency transfer and tests of fundamental physics.

NIST continues to perform research on and improve the performance of its several optical and microwave frequency standards that will be crucial to international time and frequency comparisons and ACES-mediated fundamental research: two single-aluminum ion optical standards, two single-mercury ion optical standards, two ytterbium lattice optical standards, two strontium lattice optical standards, two cesium microwave primary standards, and multiple other frequency standards under development including calcium beam optical standards.

The past project year included substantial advances in several of NIST’s frequency standards: The new NIST-F2 cesium fountain primary frequency standard was formally commissioned and evaluated. It uses a cryogenically-cooled drift tube to significantly reduce uncertainties from the blackbody radiation shift. It was demonstrated to have a fractional frequency uncertainty of approximately 1 x 10-16 (the equivalent of 1 second in 300 million years), best in the world of any microwave primary frequency standard. At the time of this report, the formal evaluation of NIST-F2 has been accepted for publication. The comparison of the NIST-F1 cesium microwave primary frequency standard operating at about 317 K compared to the NIST-F2 cesium microwave primary frequency standard operating at about 81 K also led to the world’s best measurement of the cesium blackbody radiation shift, reducing the uncertainty in that measurement by a factor of five. Since this blackbody radiation shift is generally the largest source of error in all cesium microwave primary standards, this measurement broadly enables reduced uncertainties in cesium primary standards across the world, further advancing ACES international time and frequency comparison goals.

The NIST ytterbium lattice optical clocks reported a world leading fractional frequency stability of 1.6 x 10-18 with about seven hours of averaging. This ultimate stability enables the ytterbium lattice clocks to be exceptionally precise standards for fundamental physics research. This remarkable stability also means that the ytterbium lattice clock fractional frequency stability reaches about 3 x 10-17 with only 100 seconds of averaging, dramatically advancing the goals of ACES time and frequency transfer given the relatively short visibility of the ISS to any MWL GT.

The NIST/JILA strontium lattice optical clocks reported a fractional frequency uncertainty essentially equal to the NIST ytterbium lattice clocks, and also reported a full frequency evaluation of approximately 6 x 10-18 fractional frequency uncertainty, best in the world. A full frequency evaluation of the NIST ytterbium lattice clocks will be conducted in the next several months, with comparable results expected.

At the time of this report, NIST (including NIST JILA) currently operates:

The best microwave primary frequency standard in the world (NIST-F2).

The two most stable types of frequency standards in the world, the NIST ytterbium lattice optical clocks and the NIST/JILA strontium lattice optical clocks.

The three most precise types of frequency standards in the world with formal frequency evaluations, the NIST mercury ion frequency standards, the NIST aluminum ion logic frequency standards, and the NIST/JILA strontium lattice frequency standards. The aluminum ion and strontium lattice optical standards represent the only reported frequency standards with fractional frequency uncertainties in the 10^-18 range. We fully expect the NIST ytterbium lattice optical frequency standard to demonstrate a formal frequency evaluation in the 10^-18 range in the near future.

Other laboratories across the world are expected to show similar improvements soon, and the NIST optical frequency standards are likely to continue to improve. This dramatic improvement of optical frequency standard performance strongly supports ACES goals of improved international time and frequency transfer and the ability to improve fundamental tests of physics.

NIST also continues to improve and modernize its time scale. NIST is on track to complete its major $5 million upgrade of the NIST time scale by the expected 2016 ACES launch. The improved time scale will further enhance NIST’s ability to support improved time and frequency transfer and research through ACES.

3. Theory support for ACES goals for precision time and frequency transfer and tests of fundamental physics. NIST continues to develop methods for orbit determination that are sufficiently accurate to test General Relativity with the ACES clock ensemble. To study orbit determination of the ISS, we obtained data from Johnson Space Flight Center from the SIGI2 GPS receiver on board the ISS, for March 21, 2010. The files received also included ISS sunrise and sunset

	times that we use for checking our software. The ISS orbit, as well as trajectories of radio waves used for time comparisons with clocks at the NIST ground station, are most readily analyzed in a local, freely falling, non-rotating reference frame (the "ECI" frame). GPS position and velocity are provided in the Earth-Fixed, Earth-Centered ("ECEF") reference frame. We therefore implemented the SOFA (Standards of Fundamental Astronomy) software package for performing the necessary rotations of the position and velocity data into the ECI frame for analysis. The data provided for March 21, 2010 had many gaps, probably due to poor visibility of the GPS satellite constellation. Multipath due to reflections from ISS structures is also expected to be a problem. Preliminary analysis of the data showed that characterization of one day's orbit by means of constant Keplerian orbital parameters such as semi-major axis, eccentricity, and other standard parameters, was not a good fit. Incorporation of the effect of Earth's quadrupole moment resulted in significant improvements in fitting the data; however, the fits were still too noisy. For short periods each day, ACES will be simultaneously visible from MWL GT stations at both NIST in Colorado and JPL in California. High-precision two-way time transfer will be available during such periods. This will make possible high-precision determination of ISS position and velocity through triangulation, since the baseline between JPL and NIST monitor stations will be accurately known; however, the orientation of the triangle JPL-ISS-NIST will not be completely determined by point solutions. We are developing Kalman filter software to analyse GPS data along with accurate time-transfer data. One version of this software is based on a state vector consisting of Keplerian orbital elements of the ACES position. This version has the advantage that non-Keplerian forces (e.g., due to solar radiation pressure, atmospheric drag, or similar forces). Another version of this software uses a state vector consisting of residuals of Cartesian coordinates, velocities, and accelerations, with corresponding Keplerian osculating quantities subtracted off (that is, residuals are used as state vectors). In this latter version, there are no measurements of acceleration. Instead, accelerations are computed in terms of current values of the state vector. This software is under development at NIST.
Bibliography Type:	Description: (Last Updated: 01/27/2016)
Articles in Peer-reviewed Journals	Heavner T, Donley E, Levi F, Costanzo G, Parker T, Shirley J, Ashby N, Barlow S, Jefferts S. "First accuracy evaluation of NIST-F2." Accepted for publication in Metrologia. Provisionally scheduled for April 2014. , Apr-2014
Articles in Peer-reviewed Journals	Jefferts SR, Heavner TP, Parker TE, Shirley JH, Donley EA, Ashby N, Levi F, Calonico D, Costanzo GA. "High-accuracy measurement of the black-body radiation frequency shift of the ground-state hyperfine transition in 133Cs." Phys Rev Lett. 2014 Feb 7;112(5):050801. PubMed PMID: 24580583 ; http://dx.doi.org/10.1103/PhysRevLett.112.050801 , Feb-2014
Articles in Peer-reviewed Journals	Bloom BJ, Nicholson TL, Williams JR, Campbell SL, Bishof M, Zhang X, Zhang W, Bromley SL, Ye J. "An optical lattice clock with accuracy and stability at the 10(-18) level." Nature. 2014 Feb 6;506(7486):71-5. http://dx.doi.org/10.1038/nature12941 ; PubMed PMID: 24463513 , Feb-2014
Articles in Peer-reviewed Journals	Hinkley N, Sherman JA, Phillips NB, Schioppo M, Lemke ND, Beloy K, Pizzocaro M, Oates CW, Ludlow AD. "An atomic clock with 10(-18) instability." Science. 2013 Sep 13;341(6151):1215-8. Epub 2013 Aug 22. PubMed PMID: 23970562 ; http://dx.doi.org/10.1126/science.1240420 , Sep-2013
Articles in Peer-reviewed Journals	Poli N, Oates CW, Gill P, Tino G. "Optical atomic clocks." La Rivista del Nuovo Cimento. 2013 Dec;36(12):555-624. Abstract available at: http://arxiv.org/abs/1401.2378 ; http://dx.doi.org/10.1393/ncr/i2013-10095-x , Dec-2013